

GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models

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Abstract

We investigate the potential implications of large language models (LLMs), such as Generative Pre-trained Transformers (GPTs), on the U.S. labor market, focusing on the increased capabilities arising from LLM-powered software compared to LLMs on their own. Using a new rubric, we assess occupations based on their alignment with LLM capabilities, integrating both human expertise and GPT-4 classifications. Our findings reveal that around 80% of the U.S. workforce could have at least 10% of their work tasks affected by the introduction of LLMs, while approximately 19% of workers may see at least 50% of their tasks impacted. We do not make predictions about the development or adoption timeline of such LLMs. The projected effects span all wage levels, with higher-income jobs potentially facing greater exposure to LLM capabilities and LLM-powered software. Significantly, these impacts are not restricted to industries with higher recent productivity growth. Our analysis suggests that, with access to an LLM, about 15% of all worker tasks in the US could be completed significantly faster at the same level of quality. When incorporating software and tooling built on top of LLMs, this share increases to between 47 and 56% of all tasks. This finding implies that LLM-powered software will have a substantial effect on scaling the economic impacts of the underlying models. We conclude that LLMs such as GPTs exhibit traits of general-purpose technologies, indicating that they could have considerable economic, social, and policy implications.

1 Introduction

As shown in Figure 1, recent years, months, and weeks have seen remarkable progress in the field of generative AI and large language models (LLMs). While the public often associates LLMs with various iterations of the Generative Pre-trained Transformer (GPT), LLMs can be trained using a range of architectures, and are not limited to transformer-based models (Devlin et al., 2019). LLMs can process and produce various forms of sequential data, including assembly language, protein sequences and chess games, extending beyond natural language applications alone. In this paper, we use LLMs and GPTs somewhat interchangeably, and specify in our rubric that these should be considered similar to the GPT-family of models available via ChatGPT or the OpenAI Playground (which at the time of labeling included models in the GPT-3.5 family but not in the GPT-4 family). We examine LLMs with text- and code-generating abilities, use the term "generative AI" to additionally include modalities such as images or audio, and use "LLM-powered software" to cover tools built on top of LLMs or that combine LLMs with other generative AI models.

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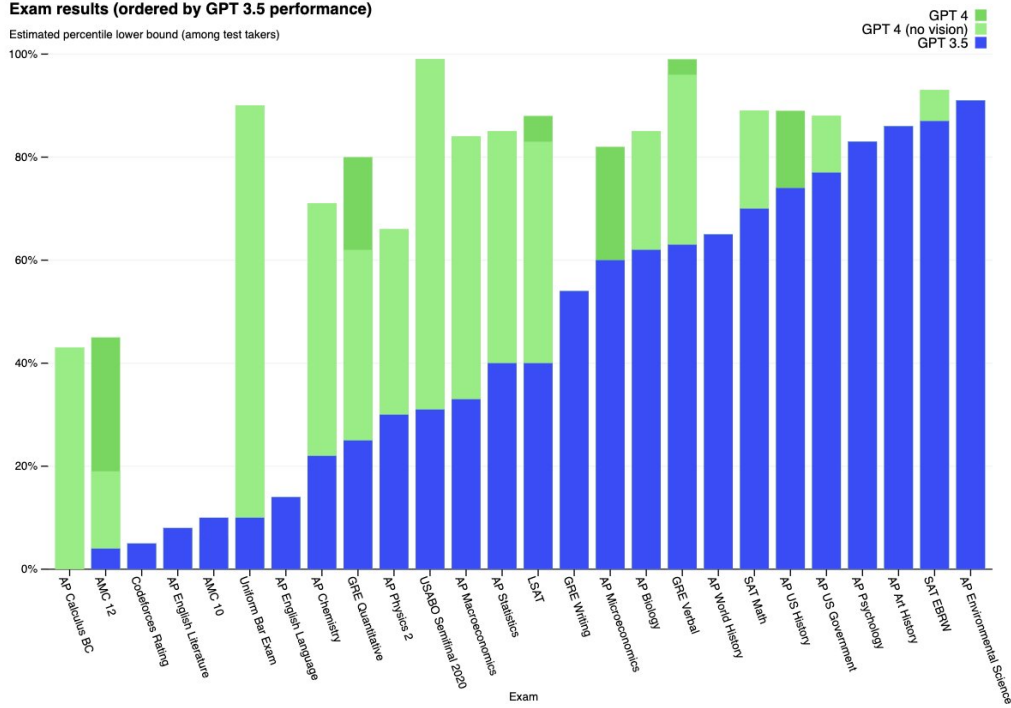


Figure 1: Taken directly from GPT-4 Technical Report (OpenAI, 2023b). To get a sense of how quickly model capabilities are progressing – consider the jump in exam performance between GPT-3.5 and GPT-4 (OpenAI, 2023b).

Our study is motivated less by the progress of these models alone though, and more by the breadth, scale, and capabilities we’ve seen in the complementary technologies developed around them. The role of complementary technologies remains to be seen, but maximizing the impact of LLMs appears contingent on integrating them with larger systems (Bresnahan, 2019; Agrawal et al., 2021). While the focus of our discussion is primarily on the generative capabilities of LLMs, it is important to note that these models can also be utilized for various tasks beyond text generation. For example, embeddings from LLMs can be used for custom search applications, and LLMs can perform tasks such as summarization and classification where the context may be largely contained in the prompt.

To complement predictions of technology’s impacts on work and provide a framework for understanding the evolving landscape of language models and their associated technologies, we propose a new rubric for assessing LLM capabilities and their potential effects on jobs. This rubric (A.1) measures the overall exposure of tasks to LLMs, following the spirit of prior work on quantifying exposure to machine learning (Brynjolfsson et al., 2018; Felten et al., 2018; Webb, 2020). We define exposure as a proxy for potential economic impact without distinguishing between labor-augmenting or labor-displacing effects. We employ human annotators and GPT-4 itself as a classifier to apply this rubric to occupational data in the U.S. economy, primarily sourced from the O*NET database.¹²

To construct our primary exposure dataset, we collected both human annotations and GPT-4 classifications, using a prompt tuned for agreement with a sample of labels from the authors. We observe similar agreement

¹This is distinct from recent social science research that makes use of LLMs to simulate human behavior (Horton, 2023; Sorensen et al., 2022)

²While our exposure rubric does not necessarily tie the concept of language models to any particular model, we were strongly motivated by our observed capabilities of GPT-4 and the suite of capabilities we saw in development with OpenAI’s launch partners (OpenAI, 2023b).